CUSTOMER CHURN ANALYSIS REPORT

1. Project Overview

A telecom company needed to understand who leaves, when they leave, and why they leave — to reduce customer churn, improve retention, and protect revenue.

"How can the company leverage customer subscription data to identify churn drivers, segment high-risk customers, and design data-backed retention strategies?"

2. Dataset Summary

- **Rows:** 7,043 customer records
- **Columns:** 21 features
- **Source:** Kaggle — Telco Customer Churn
- **Missing Data:** 11 blank values in TotalCharges column → imputed with MonthlyCharges

Feature Type	Columns
Demographics	Gender, Senior Citizen, Partner, Dependents
Subscription Info	Contract Type, Internet Service, Phone Service, Streaming
Add-on Services	Tech Support, Online Security, Online Backup, Device Protection
Billing	Monthly Charges, Total Charges, Payment Method, Paperless Billing
Behavior	Tenure (months), Churn (Yes/No)

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3. Data Cleaning & Analysis (Python)

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Missing Data Fix:** Converted blank TotalCharges strings to numeric; imputed 11 null values with MonthlyCharges.

- **Column Standardization:** Renamed all columns to snake_case for better readability and documentation.
- **Binary Encoding:** Created churn_flag column (Yes → 1, No → 0) for aggregation and analysis.
- **Feature Engineering:** Created tenure_group column — binned tenure into 6 cohorts (0-12 month, 13-24 month, 25-36 month, 37-48 month, 49-60 month, 61-72 month).
- **Cohort Analysis:** Built retention heatmap — churn rate by tenure group × contract type.
- **EDA Charts:** Created visualizations for churn by contract type, payment method, internet service, add-on services, and tenure group.
- **Revenue Calculation:** Quantified annual revenue at risk from churned customers.
- **High-Risk Segment:** Identified month-to-month + fiber optic + no tech support segment (58% churn rate).
- **Database Integration:** Exported cleaned DataFrame to MySQL database for SQL analysis.

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4. Data Analysis (SQL)

1. Overall Churn KPIs — the executive summary numbers

```
SELECT
    COUNT(customerid)                                   AS total_customers,
    SUM(churn_flag)                                     AS total_churned,
    COUNT(customerid) - SUM(churn_flag)                 AS total_retained,
    ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)   AS churn_rate_pct,
    ROUND(AVG(monthlycharges), 2)                       AS avg_monthly_charges,
    ROUND(SUM(churn_flag * monthlycharges), 2)          AS monthly_revenue_at_risk,
    ROUND(SUM(churn_flag * monthlycharges) * 12, 2)     AS annual_revenue_at_risk
FROM telco_churn_data;
```

total_customers	total_churned	total_retained	churn_rate_pct	avg_monthly_charges	monthly_revenue_at_risk	annual_revenue_at_risk
7043	1869	5174	26.54	64.76	139130.85	1669570.2

2. Churn Rate by Contract Type

```
SELECT
   contract,
   COUNT(customerid)                                        AS total_customers,
   SUM(churn_flag)                                          AS churned,
   ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)    AS churn_rate_pct,
   ROUND(AVG(monthlycharges), 2)                            AS avg_monthly_charges,
   ROUND(SUM(churn_flag * monthlycharges) * 12, 2)          AS annual_rev_at_risk
FROM telco_churn_data
GROUP BY contract
ORDER BY churn_rate_pct  DESC;
```

contract	total_customers	churned	churn_rate_pct	avg_monthly_charges	annual_rev_at_risk
Month-to-month	3875	1655	42.71	66.4	1450165.2
One year	1473	166	11.27	65.05	169421.4
Two year	1695	48	2.83	60.77	49983.6

3. Avg Monthly Charges — Churned vs Retained

```
SELECT
   churn                              AS churn_status,
   COUNT(customerid)                  AS customers,
   ROUND(AVG(monthlycharges), 2)      AS avg_monthly_charges,
   ROUND(MIN(monthlycharges), 2)      AS min_monthly_charges,
   ROUND(MAX(monthlycharges), 2)      AS max_monthly_charges,
   ROUND(AVG(totalcharges), 2)        AS avg_total_charges,
   ROUND(AVG(tenure), 1)              AS avg_tenure_months
FROM telco_churn_data
GROUP BY churn
```

ORDER BY churn DESC;

	churn_status	customers	avg_monthly_charges	min_monthly_charges	max_monthly_charges	avg_total_charges	avg_tenure_months
▶	Yes	1869	74.44	18.85	118.35	1531.8	18.0
	No	5174	61.27	18.25	118.75	2550	37.6

4. High-Risk Customer Segment

```
SELECT
   COUNT(customerid)                              AS segment_size,
   SUM(churn_flag)                                AS churned,
   ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)    AS segment_churn_rate_pct,
   ROUND(SUM(churn_flag * monthlycharges) * 12, 2)          AS annual_rev_at_risk,
   ROUND(AVG(monthlycharges), 2)                            AS avg_monthly_charges,
   -- Benchmark vs overall churn rate
   (SELECT ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)
    FROM telco_churn_data)                                  AS overall_churn_rate_pct
FROM telco_churn_data
WHERE contract       = 'Month-to-month'
  AND internetservice = 'Fiber optic'
  AND techsupport     = 'No'
  AND onlinesecurity  = 'No';
```

	segment_size	churned	segment_churn_rate_pct	annual_rev_at_risk	avg_monthly_charges	overall_churn_rate_pct
▶	1524	925	60.70	940389.6	84.65	26.54

5. Revenue at Risk

```
SELECT
   contract,
   internetservice,
   COUNT(customerid)                              AS total_customers,
   SUM(churn_flag)                                AS churned_customers,
   ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)    AS churn_rate_pct,
```

```
    ROUND(SUM(churn_flag * monthlycharges), 2)             AS monthly_rev_lost,
    ROUND(SUM(churn_flag * monthlycharges) * 12, 2)        AS annual_rev_lost
FROM telco_churn_data
GROUP BY contract, internetservice
ORDER BY annual_rev_lost DESC;
```

contract	internetservice	total_customers	churned_customers	churn_rate_pct	monthly_rev_lost	annual_rev_lost
Month-to-month	Fiber optic	2128	1162	54.61	100482	1205784
Month-to-month	DSL	1223	394	32.22	18367.25	220407
One year	Fiber optic	539	104	19.29	10571.95	126863.4
One year	DSL	570	53	9.30	3356.25	40275
Two year	Fiber optic	429	31	7.23	3246.1	38953.2
Month-to-month	No	524	99	18.89	1997.85	23974.2
Two year	DSL	628	12	1.91	805.7	9668.4
One year	No	364	9	2.47	190.25	2283
Two year	No	638	5	0.78	113.5	1362

6. Customers with >24 Months Tenure Who Still Churn

```
SELECT
    contract,
    internetservice,
    paymentmethod,
    COUNT(customerid)                          AS long_tenure_churners,
    ROUND(AVG(tenure), 1)                      AS avg_tenure_months,
    ROUND(AVG(monthlycharges), 2)              AS avg_monthly_charges,
    ROUND(AVG(totalcharges), 2)                AS avg_total_charges,
    ROUND(SUM(monthlycharges) * 12, 2)         AS annual_rev_lost
FROM telco_churn_data
WHERE tenure     > 24
  AND churn_flag = 1
GROUP BY contract, internetservice, paymentmethod
ORDER BY long_tenure_churners DESC;
```

contract	internetservice	paymentmethod	long_tenure_churners	avg_tenure_months	avg_monthly_charges	avg_total_charges	annual_rev_lost
Month-to-month	Fiber optic	Electronic check	190	40.2	93.48	3795.42	213142.2
Month-to-month	Fiber optic	Bank transfer (automatic)	55	40.9	93.26	3824.66	61549.2
One year	Fiber optic	Electronic check	49	52.4	100.88	5330.6	59316.6
Month-to-month	Fiber optic	Credit card (automatic)	48	42.3	91.02	3914.68	52426.2
One year	Fiber optic	Credit card (automatic)	23	57.6	104.02	6009.41	28708.2
One year	Fiber optic	Bank transfer (automatic)	21	55.0	102.18	5671.87	25749
Month-to-month	DSL	Electronic check	20	42.3	50.41	2116.17	12098.4
Two year	Fiber optic	Bank transfer (automatic)	15	64.3	105.34	6758.76	18961.8
Month-to-month	DSL	Credit card (automatic)	13	37.1	49.68	1885.24	7750.8
One year	DSL	Credit card (automatic)	12	48.0	70.75	3358.92	10188
Month-to-month	DSL	Bank transfer (automatic)	11	39.1	44.15	1721.96	5827.2
Month-to-month	Fiber optic	Mailed check	11	32.5	90.64	2931.46	11964
Two year	Fiber optic	Electronic check	9	58.0	101.69	5978.3	10983
One year	DSL	Mailed check	9	36.8	60.22	2135.97	6503.4
One year	DSL	Bank transfer (automatic)	7	49.3	70.27	3453.46	5902.8
One year	DSL	Electronic check	7	51.0	60.68	3067.14	5097
Two year	Fiber optic	Credit card (automatic)	6	67.0	109.2	7284.49	7862.4
One year	Fiber optic	Mailed check	5	43.4	96.83	4236.38	5809.8
Two year	DSL	Credit card (automatic)	5	63.6	72.34	4659.15	4340.4
Two year	DSL	Electronic check	4	68.5	57.78	3973.4	2773.2

7. Churn Rate by Payment Method

```
SELECT
    paymentmethod,
    COUNT(customerid)                          AS total_customers,
    SUM(churn_flag)                            AS churned,
    ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)    AS churn_rate_pct,
    ROUND(AVG(monthlycharges), 2)              AS avg_monthly_charges
FROM telco_churn_data
GROUP BY paymentmethod
ORDER BY churn_rate_pct DESC;
```

paymentmethod	total_customers	churned	churn_rate_pct	avg_monthly_charges
Electronic check	2365	1071	45.29	76.26
Mailed check	1612	308	19.11	43.92
Bank transfer (automatic)	1544	258	16.71	67.19
Credit card (automatic)	1522	232	15.24	66.51

8. Impact of Add-on Services on Churn

```
SELECT
    techsupport,
    onlinesecurity,
```

```
  COUNT(customerid)                          AS customers,
  SUM(churn_flag)                            AS churned,
  ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)   AS churn_rate_pct,
  ROUND(AVG(monthlycharges), 2)              AS avg_monthly_charges
FROM telco_churn_data
WHERE internetservice != 'No'      -- only customers with internet
GROUP BY techsupport, onlinesecurity
ORDER BY churn_rate_pct DESC;
```

techsupport	onlinesecurity	customers	churned	churn_rate_pct	avg_monthly_charges
No	No	2553	1250	48.96	74.19
Yes	No	945	211	22.33	79.76
No	Yes	920	196	21.30	75.69
Yes	Yes	1099	99	9.01	81.47

9. Churn by Tenure Group with Revenue Impact

```
SELECT
  CASE
    WHEN tenure BETWEEN  0 AND 12 THEN '0–12 mo'
    WHEN tenure BETWEEN 13 AND 24 THEN '13–24 mo'
    WHEN tenure BETWEEN 25 AND 36 THEN '25–36 mo'
    WHEN tenure BETWEEN 37 AND 48 THEN '37–48 mo'
    WHEN tenure BETWEEN 49 AND 60 THEN '49–60 mo'
    ELSE '61–72 mo'
  END                                        AS tenure_group,

  COUNT(customerid)                          AS total_customers,
  SUM(churn_flag)                            AS churned,
  ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)   AS churn_rate_pct,
  ROUND(AVG(monthlycharges), 2)              AS avg_monthly_charges,
```

```
    ROUND(SUM(churn_flag * monthlycharges) * 12, 2)          AS annual_rev_at_risk

FROM telco_churn_data
GROUP BY
    CASE
        WHEN tenure BETWEEN  0 AND 12 THEN '0–12 mo'
        WHEN tenure BETWEEN 13 AND 24 THEN '13–24 mo'
        WHEN tenure BETWEEN 25 AND 36 THEN '25–36 mo'
        WHEN tenure BETWEEN 37 AND 48 THEN '37–48 mo'
        WHEN tenure BETWEEN 49 AND 60 THEN '49–60 mo'
        ELSE '61–72 mo'
    END
ORDER BY MIN(tenure);
```

tenure_group	total_customers	churned	churn_rate_pct	avg_monthly_charges	annual_rev_at_risk
0–12 mo	2186	1037	47.44	56.1	827451
13–24 mo	1024	294	28.71	61.36	276979.8
25–36 mo	832	180	21.63	65.58	182015.4
37–48 mo	762	145	19.03	66.32	147534.6
49–60 mo	832	120	14.42	70.55	126982.8
61–72 mo	1407	93	6.61	75.95	108606.6

10. Top 3 Churning Profiles Using Window Function

```
WITH profile_churn AS (
    SELECT
        contract,
        internetservice,
        techsupport,
        paymentmethod,
        COUNT(customerid)                    AS total_customers,
```

```
        SUM(churn_flag)                    AS churned,
        ROUND(SUM(churn_flag) * 100.0 / COUNT(customerid), 2)
                                           AS churn_rate_pct,
        ROUND(SUM(churn_flag * monthlycharges) * 12, 2)     AS annual_rev_at_risk
    FROM telco_churn_data
    GROUP BY contract, internetservice, techsupport, paymentmethod
    HAVING COUNT(customerid) >= 50    -- exclude tiny segments
),
ranked AS (
    SELECT
        *,
        ROW_NUMBER() OVER (ORDER BY churn_rate_pct DESC)     AS risk_rank
    FROM profile_churn
)
SELECT
    risk_rank,
    contract,
    internetservice,
    techsupport,
    paymentmethod,
    total_customers,
    churned,
    churn_rate_pct,
    annual_rev_at_risk
FROM ranked
WHERE risk_rank <= 10
```

ORDER BY risk_rank;

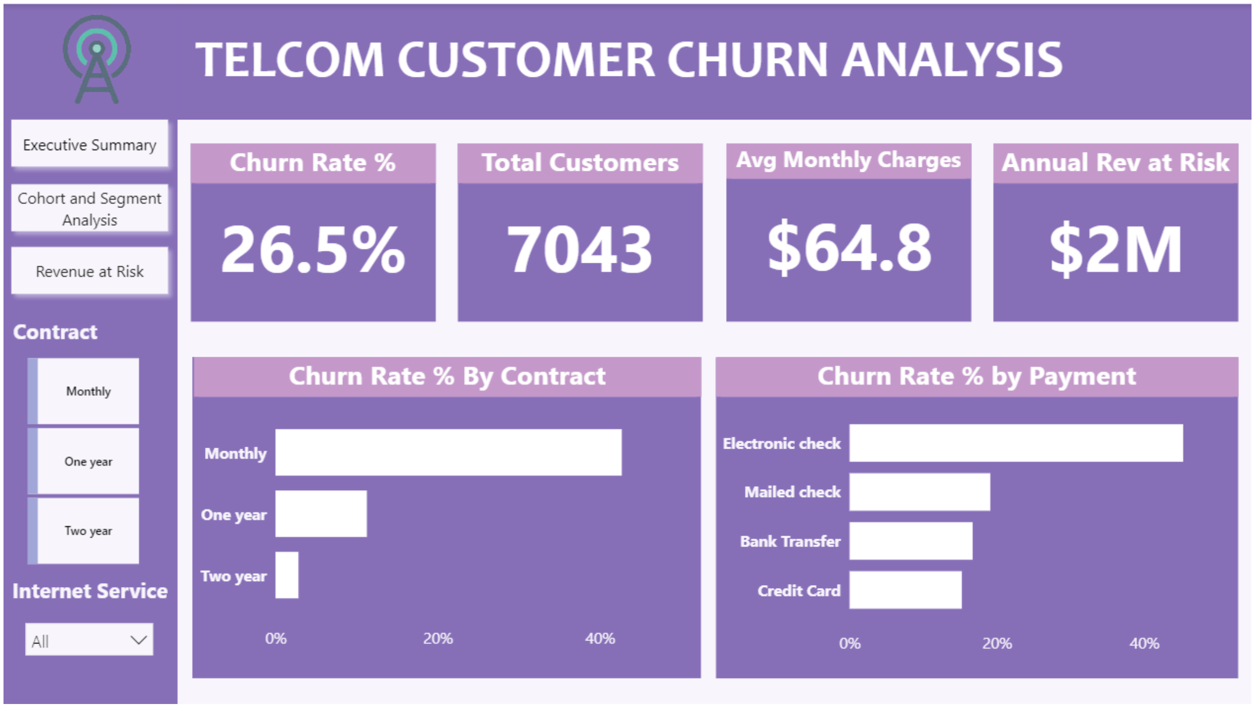
risk_rank	contract	internetservice	techsupport	paymentmethod	total_customers	churned	churn_rate_pct	annual_rev_at_risk
1	Month-to-month	Fiber optic	No	Electronic check	1138	716	62.92	736296.6
2	Month-to-month	Fiber optic	No	Mailed check	164	87	53.05	83958
3	Month-to-month	Fiber optic	No	Bank transfer (automatic)	260	127	48.85	130317.6
4	Month-to-month	DSL	No	Electronic check	365	162	44.38	85200
5	Month-to-month	Fiber optic	No	Credit card (automatic)	234	103	44.02	107416.8
6	Month-to-month	Fiber optic	Yes	Electronic check	169	73	43.20	83081.4
7	Month-to-month	DSL	No	Mailed check	262	89	33.97	48363
8	Month-to-month	Fiber optic	Yes	Bank transfer (automatic)	67	22	32.84	26427.6
9	Month-to-month	Fiber optic	Yes	Credit card (automatic)	59	19	32 32.84	21357
10	One year	Fiber optic	Yes	Electronic check	77	22	28.57	27678

5. Exploratory Insights :

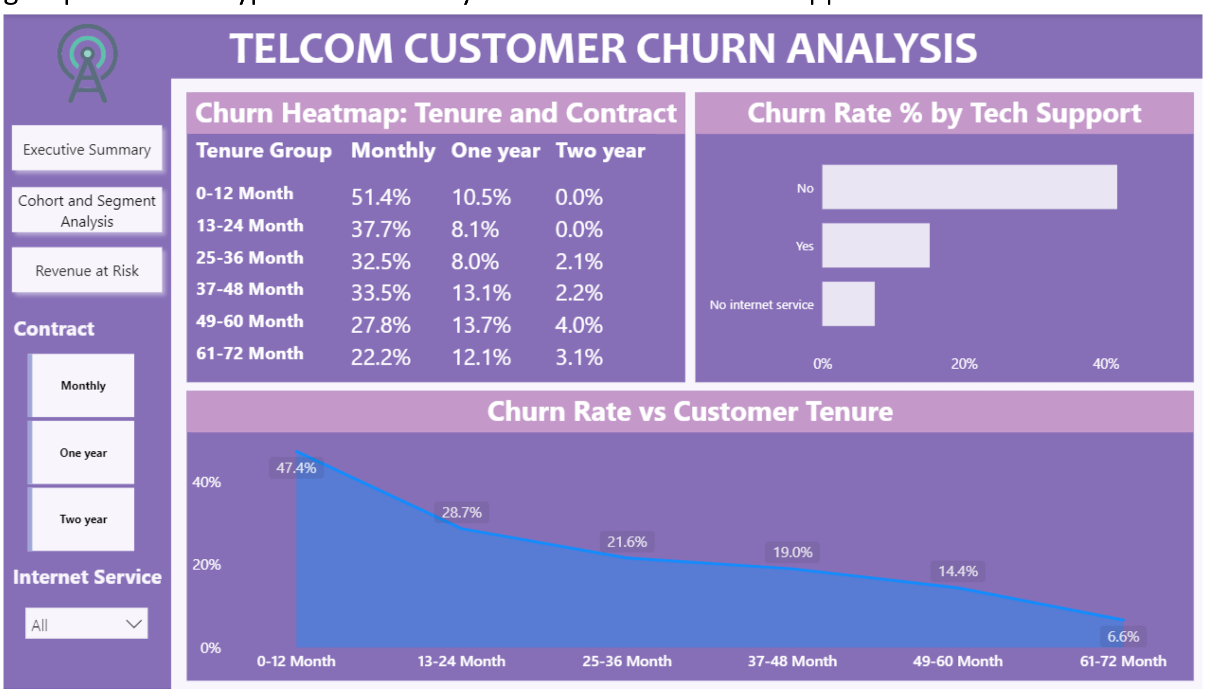
Insight	Finding
Total Customers	**7,043** customers analyzed
Overall Churn Rate	**26.5%** of customers churned
Churned Customers	**1,869** customers lost
Annual Revenue at Risk	**~$1.67M** in yearly recurring revenue at risk
Highest Risk Contract	Month-to-month — **42.7%** churn rate
Lowest Risk Contract	Two year — **2.8%** churn rate
Highest Risk Service	Fiber optic — **41.9%** churn rate
Best Retention Add-on	Tech Support + Online Security — cuts churn by **~50%**
Riskiest Payment Method	Electronic check — **45.3%** churn rate

6. Dashboard (Power BI)

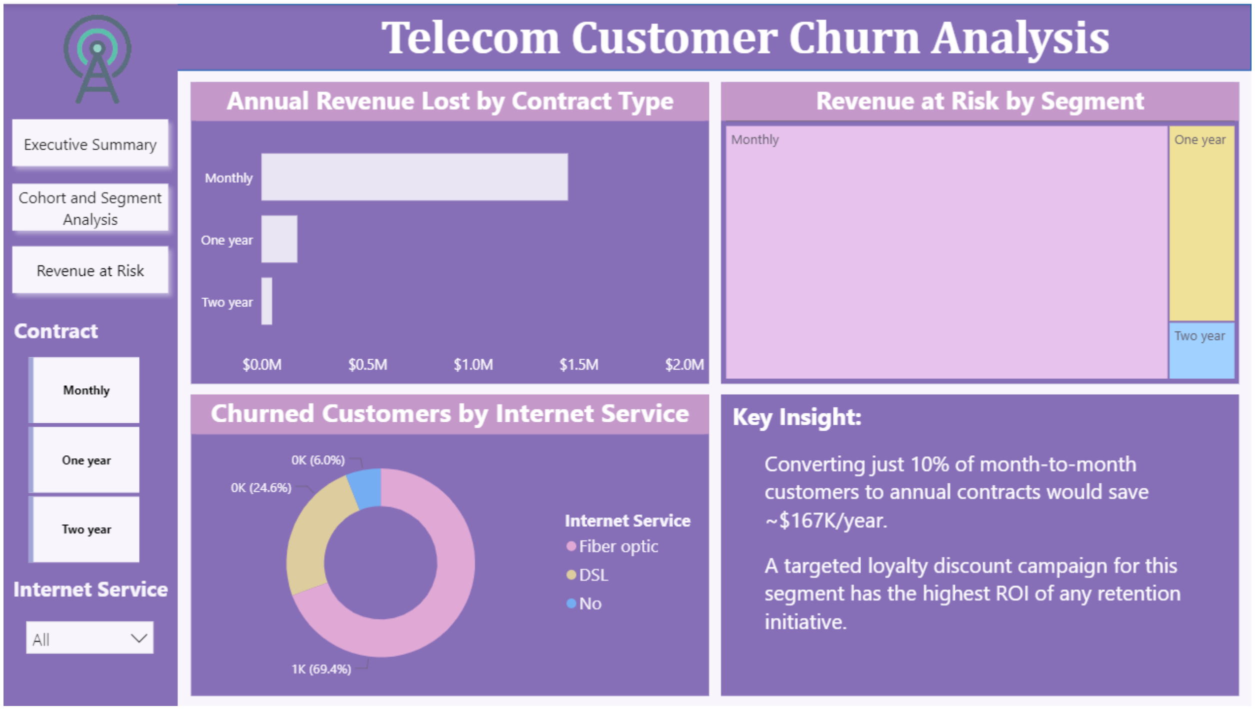
Page 1 — Executive Summary: 4 KPI cards, contract type slicer, internet service slicer, churn rate by contract type (bar chart), churn rate by payment method (bar chart).



Page 2 — Cohort & Segment Analysis: Cohort heatmap showing churn rate by tenure group × contract type. Churn rate by tenure line chart. Tech support vs churn bar chart.



Page 3 — Revenue at Risk: Annual revenue at risk by contract type (bar chart). Churned customers by internet service (donut chart). Revenue at risk treemap.



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7. Business Recommendations

1. Convert Month-to-Month Customers to Annual Contracts

Month-to-month customers churn at 42.7% vs 2.8% for two-year contracts. Offer 1-2 months free or a loyalty discount to drive annual plan upgrades. Converting just 10% of this segment saves ~$167K/year.

2. Build a 90-Day Onboarding Program

Customers in their first 12 months churn at 47.7% — the highest of any cohort. Assign dedicated onboarding support, proactive check-ins, and a free tech support trial for all new customers.

3. Bundle Tech Support + Online Security

Customers with both add-ons churn at ~15% vs ~41% for those with neither. Offer a "Protection Bundle" free for the first 3 months to fiber optic customers — the highest-risk service group.

4. Migrate Electronic Check Users to Auto-Pay

Electronic check customers churn at 45.3% vs 15-17% for auto-pay users. Offer a small monthly discount (e.g., 5%) to incentivize switching to automatic bank transfer or credit card payment.

5. Target Long-Tenure Month-to-Month Customers
2+ year customers still on month-to-month contracts are high-value retention targets. They stayed long enough to show loyalty but never upgraded. A personalized loyalty offer for this segment has the highest conversion potential.

6. Proactively Re-Engage Repeat Buyers Without Subscriptions
2,518 customers with 5+ purchases are not subscribed. A targeted re-engagement campaign — email, in-app notification, or a call — focused on this group has direct revenue upside with low acquisition cost.

This project was completed as part of a telecom customer behavior analysis initiative using Python, SQL, and Power BI.